

List of Symbols

Symbol	Description	Page
'	transpose	
$\cdot$	derivative (with respect to time)	
$\leadsto$	reachability relation	28
$\ \cdot\ _\infty$	(essential) supremum norm	56
$\sim$	indistinguishability	267
*	(subscript) Jacobian, Fréchet differential	143,470,471
*	(subscript) linearization (differential) of system	40,51
*	(superscript) conjugate transpose; adjoint operator	91,107
*	convolution, only in Chapter 7	330
#	pseudoinverse of operator or matrix	109,458
$\diamond$	empty (i.e., length zero) control	26
∇	gradient	143
$[\cdot, \cdot]$	Lie bracket of vector fields	144
ad_f	ad operator (Lie bracket)	144
α	map from initial state and control to final state	57
A	state matrix for linear systems	37,47,94
Ad_{tX}	“Ad” operator on vector fields (conjugation by e^{tX})	167
B	input matrix for linear systems	37,47,94
$\mathcal{B}_\rho(x)$	open ball of radius ρ about x	125
$\overline{\mathcal{B}}_\rho(x)$	closed ball of radius ρ about x	125,480
$\mathbf{B}_{n,m}$	class of B matrices for recurrent nets	131
co	convex hull	434
$\mathbb{C}$	complexes	
C	output matrix for linear systems	37,47
$\mathcal{C}(x)$	states that can be controlled to x	89
$\mathcal{C}^T(x)$	states controllable to x in time exactly T	89
$\mathcal{C}^{\leq T}(x)$	states controllable to x in time $\leq T$	158

$\mathcal{C}_{\mathcal{V}}^{\leq T}(x)$	states controllable to x in time $\leq T$ without leaving $\mathcal{V}$	158
$\mathcal{C}^1$	continuously differentiable functions	471
$\mathcal{C}^\infty$	smooth (infinitely differentiable) functions	469
$\mathcal{C}^k$	functions k -times continuously differentiable	469
$\mathcal{C}_n^0(\sigma, \tau)$	continuous maps on $[\sigma, \tau]$	480
$\mathcal{C}^\omega$	analytic functions	492
χ	characteristic polynomial	91
$\mathcal{D}$	domain of dynamics (continuous-time)	44
$\mathcal{D}_{\sigma, \tau}$	domain of transition map, fixed interval $[\sigma, \tau]$	53
$\mathcal{D}_f$	domain of flow associated to vector field f	149
$\mathcal{D}_\phi$	domain of transition map	26
$\mathcal{D}_\lambda$	domain of response map	30
δ	sampling interval	7, 72
Δ	distribution (in the sense of differential geometry)	171
e^{tA}	exponential of matrix tA	496
e^A	exponential of matrix A	496
$e^{tf}x^0$	alternative notation for flow $\phi(t, x^0)$	152
$f(t, x, u)$	right-hand side for continuous-time system	41
$f(x, u)$	rhs for continuous-time time-invariant system	41
f, g	vector fields (typically associated to controls)	143
f_u	vector field associated to input value u	156
ϕ	transition (global dynamics) map	26
$\mathbb{F}$	smooth functions	143
Φ	fundamental matrix, for linear differential equation	496
Γ	trajectory: pair $(\xi(\cdot), \omega(\cdot))$	28
h	output map	27, 33
$\mathcal{H}$	Hankel matrix	289
$\text{int}(V)$	interior of set V	
I	identity matrix	
I	in Section 6.2, indistinguishability relation	273
$\mathcal{I}$	interval in $\mathbb{R}$ or relative to $\mathcal{T}$	25
φ	smooth function (nonlinear controllability chapter)	143
$\mathcal{J}$	total cost in optimal control problems	354, 367
$k(t)$	kernel of integral operator	107
$k(x)$	feedback	238
κ_i	Kronecker indexes	192
$\mathbb{K}$	field; for continuous-time systems $= \mathbb{R}$ or $\mathbb{C}$	36

$K(t, s)$	kernel of integral operator	50
$\{\cdot\}_{\text{LA}}$	Lie algebra generated	145
L_f	directional (Lie) derivative	144
$\mathcal{L}^q$	q -integrable functions: φ so that $\int \varphi ^q < \infty$	
$\mathcal{L}_r^2$	$(\mathcal{L}^2)^r$, or square-integrable functions $\rightarrow \mathbb{K}^r$	107
$\mathcal{L}_m^1$	$(\mathcal{L}^1)^m$, or integrable functions $\rightarrow \mathbb{K}^r$	434
$\mathcal{L}_{\mathcal{V}}^{\infty}(\mathcal{I})$	essentially bounded maps $\mathcal{I} \rightarrow$ to metric space $\mathcal{V}$	479
$\mathcal{L}_{\mathcal{V}}^{\infty, \text{loc}}(\mathcal{I})$	locally essentially bounded maps $\mathcal{I} \rightarrow \mathcal{V}$	479
$\mathcal{L}_{\mathcal{V}}^{\infty}, \mathcal{L}_{\mathcal{V}}^{\infty, \text{loc}}$	same as $\mathcal{L}_{\mathcal{V}}^{\infty}(\mathcal{I})$, $\mathcal{L}_{\mathcal{V}}^{\infty, \text{loc}}(\mathcal{I})$ if $\mathcal{I}$ clear from context	479
$\mathcal{L}_r^{\infty}, \mathcal{L}_r^{\infty, \text{loc}}$	same as $\mathcal{L}_{\mathcal{V}}^{\infty}$, $\mathcal{L}_{\mathcal{V}}^{\infty, \text{loc}}$, if $\mathcal{V} = \mathbb{R}^r$ (or $\mathbb{C}^r$)	479
$\mathcal{L}_{\mathcal{U}}$	used in Chapter 10 instead of $\mathcal{L}_{\mathcal{U}}^{\infty}$ ($\mathcal{U}$ is compact)	429
Λ	input-output behavior specification	30
λ	input-output behavior map	30
$\lambda^{\sigma, \tau}$	λ for fixed interval, just function of input	30
$\bar{\lambda}$	entire output function (λ is just last output value)	31
λ_{Σ, x^0}	behavior associated to a system Σ and initial state x^0	265
λ_{x^0}	same as λ_{Σ, x^0} , if Σ clear from context	265
$\lambda_{x^0}^{\sigma, \tau}$	same as λ_{x^0} , but fixing time interval	266
m	dimension of input-value space	37,47
n	dimension of state space	37,47
$\mathbf{0}$	input identically $\equiv 0$	38
$\mathcal{O}$	in Chapters 4 and 10, open subset of Euclidean space	143
$\mathcal{O}$	in Chapters 6 and 7, states unobservable from zero	274
$\mathbf{O}_k$	observability matrix (has size $kp \times n$)	275
p	dimension of output-value space	37,47
p	cost on terminal states	354
π	coordinate functions	144
ψ	map that assigns state path	28,57
$\mathcal{P}$	transition map (discrete-time)	33
q	instantaneous trajectory cost	354
$\mathcal{Q}$	total trajectory cost	354
Q	integral cost on states, in LQ problems	367
$\mathbb{R}$	reals	
$\mathbb{R}_{\geq 0}, \mathbb{R}_+$	nonnegative reals	
$\mathbf{R}$	reachability matrix	90
$\mathbf{R}_k$	reachability matrix in k steps (has size $n \times kn$)	289
$\mathbf{R}(A, B)$	reachability matrix for pair	94

$\mathcal{R}(A, B)$	reachability space for pair	94
R	integral cost on controls, in LQ problems	367
$\mathcal{R}(x)$	reachable set from state x	86
$\mathcal{R}^T(x)$	reachable set from x in time exactly T	86
$\mathcal{R}^{\leq T}(x)$	reachable set from x in time $\leq T$	158
$\mathcal{R}_{\mathcal{V}}^{\leq T}(x)$	reachable set from x in time $\leq T$ without leaving $\mathcal{V}$	158
S	cost on final states, in LQ problems	367
$\mathcal{S}_{n,m}$	pairs $A \in \mathbb{K}^{n \times n}, B \in \mathbb{K}^{n \times m}$	98
$\mathcal{S}_{n,m}^c$	controllable pairs in $\mathcal{S}_{n,m}$	98
σ	time, usually initial	
$\sigma_{\min}$	smallest singular value	456
Σ	system	26
$\Sigma_{[\delta]}$	δ -sampled continuous-time system	72
τ	time, usually final	26
$\tau_{\min}, \tau_{\max}$	minimal and maximal time for solution	487
$\mathcal{T}$	time set	26
u	input value also input <i>functions</i> in informal discussions (instead of ω)	33,41
$\mathcal{U}$	input-value space	26
$\mathcal{U}^{[\sigma, \tau]}$	controls $[\sigma, \tau] \rightarrow \mathcal{U}$ (for continuous time, same as $(\sigma, \tau) \rightarrow \mathcal{U}$ or $[\sigma, \tau] \rightarrow \mathcal{U}$)	26
$\mathbb{V}$	smooth vector fields	143
V	(control-) Lyapunov function	221
$\xrightarrow{w}$	weak convergence	430
ω	input (control) functions	26
x	state also state <i>functions</i> in informal discussions (instead of ξ)	41,26,33
x^0	state, usually initial, or equilibrium	31
x^f	final state (in optimal control problems, typically)	429
$\mathcal{X}$	state space	26
X, Y, Z	vector fields (typically in generated Lie Algebra)	145
ξ	state path	28
y	output value	27,33
$\mathcal{Y}$	output-value space	27
$\mathbb{Z}$	integers	